

Calculus 0314

Quiz 1.

- (1) Under ideal conditions a certain bacteria population is known to double every three hours.

Suppose that there are initially 100 bacteria. What is the size of the population after t hours?

(6%) $100 \times (2)^{t/3}$

- (2) (a) Find the inverse function of $f(x) = \frac{4x-1}{2x+3}$. (4%) $f^{-1}(x) = \frac{3x+1}{-2x+4}$

(b) What is the range of f ? (4%) $\mathbb{R} \setminus \{2\}$

- (3) Solve for $\ln x + \ln(x-1) = 1$. (6%) $x = \frac{1+\sqrt{1+4e}}{2}$

- (4) $\tan^{-1}(\tan \frac{42}{3}) =$ (6%) $14 - 4\pi$

- (5) Determine the limit.

(a) $\lim_{h \rightarrow 0} \frac{(4+h)^2 - 16}{h}$. (6%) 8

(b) $\lim_{t \rightarrow 0} (\frac{1}{t\sqrt{1+t}} - \frac{1}{t})$. (6%) $-\frac{1}{2}$

(c) $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin \frac{\pi}{x}}$. (6%) 0

(d) $\lim_{x \rightarrow -2^-} [x]$, where $[x]$ is the greatest integer less than or equal to x . (6%) -3

(e) $\lim_{x \rightarrow 0} \frac{|2x-1| - |2x+1|}{x}$. (6%) -4

- (6) If $\lim_{x \rightarrow 1} \frac{3x^2 + ax + a + 3}{x^2 + x - 2}$ exists, then

(a) find a . (4%) -3

(b) find the limit. (4%) 1

- (7) Find numbers a and b such that $\lim_{x \rightarrow 0} \frac{\sqrt{ax+b}-2}{x} = 1$. (6%) $a = b = 4$

- (8) A machinist is required to manufacture a circular metal disk with area $100 \pi \text{ cm}^2$.

(a) What radius produces such a disk? (6%) 10 cm

(b) Let the following information be known. If the machinist is allowed an error tolerance

of $\pm \pi \text{ cm}^2$ in the area of the disk, then the machinist must control the radius within an

error tolerance of $\pm 0.049 \text{ cm}$. In terms of the ε, δ definition of $\lim_{x \rightarrow a} f(x) = L$, (i) what is

x ? (ii) What is $f(x)$? (iii) What is a ? (iv) What is L ? (v) What value of ε is given? (vi)

What is the corresponding value of δ ? (9%) (i) radius (ii) $f(x) = \pi x^2$ (iii) 10 (iv) 100π

(v) π (vi) 0.049

- (9) (a) State the definition of $\lim_{x \rightarrow a} f(x) = L$. (7%)

(b) Use the definition to prove $\lim_{x \rightarrow 1} (2x+3) = 5$. (8%)